

Anchored Recovery of the Dyadic Uniform Spectrum

A sharp stretched-exponential modulus,
lacunary counterexamples to Hölder stability,
and Lipschitz recovery of every fixed prefix

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Abstract

Let μ_a be the law of $\sum_{j \geq 1} a_j U_j$, where the U_j are independent uniforms on $[-1, 1]$ and a is a decreasing summable sequence. We resolve the anchored dyadic recovery question left in the ProveIt manuscript *Recovering Uniform Factors*: one of the two laws being compared is now exactly the Rvachev law μ_0 , with spectrum $a_j^0 = 2^{-j}$. For either total variation or Kolmogorov distance, the anchored inverse modulus has the scale

$$\omega(\varepsilon) = \exp \left[-\Theta \left(\sqrt{\log(1/\varepsilon)} \right) \right].$$

Thus no positive Hölder exponent is possible, but recovery at the reference law is faster than every negative power of $\log(1/\varepsilon)$. The lower bound holds in the predecessor's exact fixed-variance, uniformly smooth class, and even after imposing $a_{j+1} \leq \rho a_j$ for any fixed $\rho > 1/2$. In contrast, every fixed leading prefix is locally Lipschitz-stable, with the rest of the spectrum completely unknown subject to a support bound. The upper proof reconstructs primitive factor periods from stable zero counts by divisor inversion. The lower proof perturbs the constant coefficient of a polynomial with geometrically spaced positive roots, matches many cumulants, and separates low-frequency cancellation from high-frequency smoothing. Explicit finite-error formulas, a testing sample-complexity theorem, exact rational regression tests, a proof audit, and further questions are included. The exponent $1/2$ is sharp; the leading constant in the stretched exponential is not determined.

Status. This is a proposed research contribution supported by proofs written below, not a Lean- or Rocq-checked result. The repository question and the comparison class are identified precisely. Literature priority has not been established exhaustively, and independent review remains necessary.

Contents

1	The question, the comparison class, and the contribution	3
1.1	An anchored question that pairwise instability does not answer	3
1.2	Definitions	3
1.3	Main results	4
1.4	What is new here, and what is established background	5
2	Analytic and algebraic preliminaries	5
2.1	Entire transforms and their zeros	5
2.2	Two measurement norms control the same contours	6
2.3	Power sums and the logarithm of sinc	6
3	Recovering primitive periods from perturbed zero counts	6
4	An explicit upper modulus and the prefix theorem	7
4.1	A uniform lower bound away from the zero lattice	7
4.2	Why multiple transform zeros do not destroy prefix Lipschitzness	8
5	A positive-root perturbation of existing dyadic factors	9
5.1	A constant-coefficient perturbation with controlled real roots	9
5.2	The full infinite spectrum	10
5.3	Membership in the exact smooth class	11
6	A quantitative bound against the exact reference law	12
6.1	The unchanged odd-indexed background	12
6.2	An explicit low-frequency estimate	13
6.3	The full-law error and its quadratic logarithm	14
7	Completion of the anchored theorem and its interpretation	15
7.1	Exactly which repository questions are answered	15
7.2	Why the two exponents are geometrically natural	16
8	Testing complexity and honest confidence sets	16
8.1	A testing formulation that is genuinely informative	16
8.2	Confidence sets that shrink at the reference	17
9	Reproducible checks and finite-error scales	17
9.1	Exact arithmetic, diagnostics, and what each establishes	17
9.2	Scale of the anchored witnesses	18
9.3	Scale of a finite recovery certificate	18
9.4	How to reproduce the package	19
10	Proof dependencies and repository provenance	19
10.1	Claim-level dependency audit	19
10.2	Inspected source and exact question mapping	19
10.3	A concrete formalization route	20
11	Further research questions	21
12	Conclusion	22

1 The question, the comparison class, and the contribution

1.1 An anchored question that pairwise instability does not answer

The ProveIt Fabius-function research collection contains the manuscript *Recovering Uniform Factors: Exact Moment Fibres and Logarithmic Instability Near the Fabius–Rvachev Law* [1]. Its second research question asks whether a power-law inverse estimate might hold relative to the *single exact dyadic reference spectrum*. The fourth asks whether geometric separation improves recovery, and the fifth asks about recovering only a fixed number of leading factors. These are logically distinct from its pairwise construction: two laws can be extremely close to each other and both approach a reference law without either being comparably close to that reference law.

Here every lower-bound comparison is directly against the exact law μ_0 . Moreover, the perturbations do not insert a cluster of almost equal scales. They change finitely many existing even-indexed dyadic scales by small relative amounts and retain the odd-indexed scales exactly. Consequently they remain uniformly geometrically separated. This is the main structural difference from the predecessor’s construction.

The inspected repository reference is

39472967ce67566d7c530fd5b8d6a3a95e68fe22.

The exact source path and the relevant question locations are recorded in section 10. The repository itself labels the predecessor as an unreviewed archival arrival. We do not use its quantitative theorems as unproved inputs: the present construction and both analytic bounds are proved independently below. Its class definition is retained deliberately, so that the new obstruction cannot be dismissed as a change to rougher densities.

1.2 Definitions

For $L > 0$ let

$$\mathcal{A}_L = \left\{ a_1 \geq a_2 \geq \cdots \geq 0 : \sum_{j \geq 1} a_j \leq L \right\}, \quad X_a = \sum_{j \geq 1} a_j U_j, \quad \mu_a = \text{Law}(X_a). \quad (1.1)$$

The series converges absolutely for every outcome. Zero entries are padding; they are not extra identifiable factors. Define

$$d_\infty(a, b) = \sup_{j \geq 1} |a_j - b_j|, \quad a^0 = (2^{-1}, 2^{-2}, \dots), \quad \mu_0 = \mu_{a^0}. \quad (1.2)$$

We write $f_0 = \text{up}$ for the density of μ_0 . With this convention, $\text{supp } \mu_0 = [-1, 1]$, $\text{Var}(X_{a^0}) = 1/9$, and $f_0(0) = 1$.

For probability measures let

$$\text{TV}(\mu, \nu) = \sup_B |\mu(B) - \nu(B)|, \quad (1.3)$$

$$d_K(\mu, \nu) = \sup_{x \in \mathbb{R}} |F_\mu(x) - F_\nu(x)|. \quad (1.4)$$

In particular $d_K \leq \text{TV}$, and for densities total variation is one-half the L^1 distance. Every logarithm is natural unless a base is displayed.

Definition 1.1 (The predecessor's smooth class). Let $\mathcal{K} \subset \mathcal{A}_3$ consist of spectra whose laws have variance $1/9$ and a density $f_a \in C_c^\infty(\mathbb{R})$ satisfying

$$\|f_a^{(r)}\|_\infty \leq 2^{r+1} \|f_0^{(r)}\|_\infty \quad (r = 0, 1, 2, \dots). \quad (1.5)$$

For $1/2 < \rho < 1$ put

$$\mathcal{K}_\rho = \{a \in \mathcal{K} : a_{j+1} \leq \rho a_j \text{ for every } j\}.$$

Both classes contain a^0 .

Definition 1.2 (Anchored inverse modulus). For a class $\mathcal{C}$ containing a^0 and a law distance D , set

$$\omega_{\mathcal{C},D}(\varepsilon) = \sup\{d_\infty(a, a^0) : a \in \mathcal{C}, D(\mu_a, \mu_0) \leq \varepsilon\}. \quad (1.6)$$

This is an anchored modulus, not a supremum over two freely varying spectra.

1.3 Main results

Theorem 1.3 (Sharp anchored scale). Take $D = \text{TV}$ or $D = d_K$. For each of the classes

$$\mathcal{C} = \mathcal{A}_L \ (L > 1), \quad \mathcal{C} = \mathcal{K}, \quad \text{or} \quad \mathcal{C} = \mathcal{K}_\rho \ (1/2 < \rho < 1),$$

there are positive constants c, C, ε_0 such that

$$\exp\left[-C\sqrt{\log(1/\varepsilon)}\right] \leq \omega_{\mathcal{C},D}(\varepsilon) \leq \exp\left[-c\sqrt{\log(1/\varepsilon)}\right] \quad (0 < \varepsilon < \varepsilon_0). \quad (1.7)$$

More quantitatively, the proofs give

$$\sqrt{\log 2} \leq \liminf_{\varepsilon \downarrow 0} \frac{\log(1/\omega_{\mathcal{C},D}(\varepsilon))}{\sqrt{\log(1/\varepsilon)}} \leq \limsup_{\varepsilon \downarrow 0} \frac{\log(1/\omega_{\mathcal{C},D}(\varepsilon))}{\sqrt{\log(1/\varepsilon)}} \leq \frac{20}{3} \sqrt{\log 2}. \quad (1.8)$$

No assertion that the two limiting constants coincide is made.

Theorem 1.4 (Fixed prefixes are Lipschitz-stable). For every $L \geq 1$ and every fixed integer $r \geq 1$ there are $C_{L,r} < \infty$ and $\varepsilon_{L,r} > 0$ such that

$$\max_{1 \leq j \leq r} |a_j - 2^{-j}| \leq C_{L,r} d_K(\mu_a, \mu_0) \quad \text{if } a \in \mathcal{A}_L, \quad d_K(\mu_a, \mu_0) \leq \varepsilon_{L,r}. \quad (1.9)$$

The same is true with total variation. No tail separation, fixed tail, or finite-capacity assumption is required. One can take $\log C_{L,r} \leq (\log 2)r^2 + O_L(r+1)$.

There is no contradiction. A fixed prefix and the whole infinite spectrum are different losses. The constants in the prefix theorem deteriorate rapidly as the prefix grows; the least detectable perturbed coordinate in the lower-bound construction also moves to infinity.

The statistical consequence is likewise stated in a form that avoids a pointwise-estimation trap. Testing the fixed null $a = a^0$ against $d_\infty(a, a^0) \geq \delta$ has sample complexity

$$N_*(\delta) = \exp\left[\Theta\left(\log^2(1/\delta)\right)\right] \quad (1.10)$$

for fixed nontrivial error levels. We prove this for alternatives in $\mathcal{K}_\rho$ and give an upper test valid on all of $\mathcal{A}_3$.

1.4 What is new here, and what is established background

Qualitative identifiability of generalized uniform sums is established work: Billey and Swanson give a unique decreasing-sequence parametrization and a topological theory that also permits a Gaussian component [2]. Arias de Reyna is a primary source for the compactly supported smooth function and its transform conventions [3]. The use of Newton identities, Rouché's theorem, Plancherel's theorem, and empirical distribution-function bounds is classical.

The proposed contribution is the specific quantitative combination: an anchored stretched-exponential modulus with matching exponent; a counterexample family inside the exact smooth class and within any separation class $\rho > 1/2$; and Lipschitz recovery of a fixed prefix despite an unrestricted infinite tail. These are not presented as new versions of identifiability itself. Nor does this article settle the predecessor's *pairwise* optimal logarithmic modulus, its exact-support-endpoint question, or general minimax estimation over all spectra.

2 Analytic and algebraic preliminaries

2.1 Entire transforms and their zeros

Lemma 2.1. *For $a \in \mathcal{A}_L$ the characteristic function extends to the entire function*

$$\phi_a(z) = \mathbb{E}e^{izX_a} = \prod_{j \geq 1} \text{sinc}(a_j z), \quad \text{sinc } z = \frac{\sin z}{z}, \quad \text{sinc } 0 = 1. \quad (2.1)$$

Its positive zeros, with multiplicity, are exactly

$$\left\{ \frac{k\pi}{a_j} : a_j > 0, \ k \geq 1 \right\}. \quad (2.2)$$

In particular it has no nonreal zeros and no extra zeros created by the infinite product.

Proof. Since $|X_a| \leq L$, the expectation is entire by locally uniform domination. Independence gives the finite products, and absolute convergence of the random series gives their locally uniform convergence to the expectation. Alternatively, $\sum_j |\text{sinc}(a_j z) - 1|$ converges uniformly on every compact set because $\sum a_j^2 < \infty$. After removing finitely many initial factors, every remaining factor is close to one and its analytic logarithm is summable. Thus the tail product is nonzero at every point not already a factor zero. Only finitely many factors can vanish at a given nonzero point, because such a factor must have $a_j \geq \pi/|z|$. This proves the multiplicity assertion. $\square$

For the dyadic law put

$$\Phi_0(w) = \phi_0(2\pi w) = \prod_{j \geq 1} \text{sinc}(\pi 2^{1-j} w). \quad (2.3)$$

Its positive zero at the integer m has multiplicity

$$c_m = 1 + v_2(m). \quad (2.4)$$

Indeed the contributing factors are precisely those with $2^{j-1} \mid m$.

2.2 Two measurement norms control the same contours

Lemma 2.2. *Suppose μ, ν are supported in $[-L, L]$. Then*

$$|\phi_\mu(z) - \phi_\nu(z)| \leq 2e^{L|\operatorname{Im} z|} \operatorname{TV}(\mu, \nu), \quad (2.5)$$

$$|\phi_\mu(z) - \phi_\nu(z)| \leq 2L|z|e^{L|\operatorname{Im} z|} d_K(\mu, \nu). \quad (2.6)$$

Proof. The first inequality follows by integrating e^{izx} against the signed measure $\mu - \nu$, whose total mass variation is $2 \operatorname{TV}(\mu, \nu)$. For the second, Stieltjes integration by parts gives

$$\phi_\mu(z) - \phi_\nu(z) = -iz \int_{-L}^L e^{izx} (F_\mu(x) - F_\nu(x)) dx.$$

One may first integrate over a slightly larger interval, where the boundary terms vanish, so atoms at the endpoints cause no difficulty. Taking absolute values gives the claim. $\square$

2.3 Power sums and the logarithm of sinc

For $|u| < \pi$, Euler's sine product and absolute summation give

$$\log \operatorname{sinc} u = - \sum_{k \geq 1} \frac{\zeta(2k)}{k\pi^{2k}} u^{2k}. \quad (2.7)$$

For real $|u| < \pi$ this is the real logarithm of a positive number. In the complex disk it is the analytic branch vanishing at zero. Thus equality of $\sum a_j^{2k}$ for $1 \leq k \leq m$ implies equality of cumulants, and hence moments, through degree $2m + 1$, since all odd cumulants vanish. We shall use this only for finite blocks, where all algebraic manipulations are immediate.

3 Recovering primitive periods from perturbed zero counts

The crucial discrete step is not generic stability of a multiple root. It uses the fact that each uniform factor contributes *an entire harmonic progression* of zeros.

Lemma 3.1 (Stable divisor reconstruction). *Fix $\eta = 1/10$ and an integer $M \geq 4$. Let $0 < \beta_1 \leq \beta_2 \leq \dots$ be a finite or locally finite sequence, with $\beta_i \rightarrow \infty$ in the infinite case. Suppose the multiset $\{k\beta_i : k \geq 1\}$ has the following properties up to $2M + \eta$: all its points lie in the intervals*

$$I_m = (m - \eta, m + \eta), \quad 1 \leq m \leq 2M,$$

and exactly $1 + v_2(m)$ points, counting multiplicity, lie in I_m . Then for every power of two $d \leq M$ there is exactly one primitive period $\beta(d) \in I_d$, and there are no other primitive periods at most $M + \eta$. Moreover,

$$|\beta(d) - d| \leq \frac{\eta\beta(d)}{M}. \quad (3.1)$$

If $b_i = 1/(2\beta_i)$, decreasingly ordered and padded by zeros when necessary, then

$$d_\infty(b, a^0) \leq \frac{1}{2M}. \quad (3.2)$$

Proof. Any primitive period $\beta \leq M + \eta$ itself is an observed zero, so it lies in a unique I_d , $1 \leq d \leq M$. Write $\beta = d + e$, $|e| < \eta$. For all k with $k\beta \leq 2M + \eta$, let m_k be the integer satisfying $|k\beta - m_k| < \eta$. Taking $m_0 = 0$, consecutive integers satisfy

$$|(m_k - m_{k-1}) - d| < 3\eta < 1.$$

Therefore $m_k - m_{k-1} = d$, and induction gives

$$m_k = kd, \quad |ke| < \eta. \quad (3.3)$$

This prevents a progression from gradually drifting to a different lattice class. Merely locating its first zero would not give this conclusion.

If $kd \leq M$, then $k\beta \leq M + M\eta < 2M + \eta$, so this zero is indeed observed and belongs to I_{kd} . Conversely, a zero in I_m for $m \leq M$ comes from a primitive period at most $M + \eta$, and (3.3) shows that its associated integer d divides m . Writing n_d for the number of primitive periods in I_d , we obtain

$$1 + v_2(m) = \sum_{d|m} n_d \quad (1 \leq m \leq M). \quad (3.4)$$

This triangular system has a unique solution. The solution is $n_d = 1$ when d is a power of two and $n_d = 0$ otherwise, because a positive integer m has exactly $1 + v_2(m)$ power-of-two divisors.

For a recovered period put $K = \lfloor (2M + \eta)/\beta \rfloor$. Since $\beta \leq M + \eta$, we have $K \geq M/\beta$. Equation (3.3) yields $|e| \leq \eta/K \leq \eta\beta/M$. Consequently

$$\left| \frac{1}{2\beta(d)} - \frac{1}{2d} \right| \leq \frac{\eta}{2Md}.$$

These are exactly the leading dyadic factors with $d \leq M$. Every remaining period is greater than $M + \eta$, while the next dyadic primitive period is greater than M . The two nonnegative tails are therefore bounded above by $1/(2M)$. This proves (3.2). $\square$

Remark 3.2. The lemma is an approximate-lattice statement with exact multiplicities. It is robust to multiplicities in the unknown spectrum and makes no a priori separation assumption on that spectrum. Divisor inversion forces the relevant separation after the zero counts have been established.

4 An explicit upper modulus and the prefix theorem

4.1 A uniform lower bound away from the zero lattice

Keep $\eta = 1/10$ and define, for $M \geq 4$,

$$r_0 = 2\pi\eta, \quad R_M = 2\pi(2M + \eta), \quad J_M = \lceil \log_2(2R_M) \rceil, \quad (4.1)$$

$$B_M = e^{-1} \left(\frac{4\eta}{R_M} \right)^{J_M}, \quad \varepsilon_M(L) = \frac{B_M}{4(1 + LR_M)e^{Lr_0}}. \quad (4.2)$$

Lemma 4.1. *The bound $|\phi_0(z)| \geq B_M$ holds on every circle $|z - 2\pi m| = r_0$, $1 \leq m \leq 2M$, and at every real $z \in [r_0, R_M]$ outside the corresponding open disks. For real $0 \leq z \leq r_0$, one has $\phi_0(z) > 1/2$.*

Proof. For every complex u ,

$$|\sin u|^2 = \sin^2(\operatorname{Re} u) + \sinh^2(\operatorname{Im} u) \geq \frac{4}{\pi^2} \operatorname{dist}(u, \pi\mathbb{Z})^2.$$

Here one reduces the real part to $[-\pi/2, \pi/2]$, uses $|\sin x| \geq 2|x|/\pi$, and uses $|\sinh y| \geq |y|$. At any point specified in the lemma, $\operatorname{dist}(z, 2\pi\mathbb{Z}) \geq r_0$. Since $2^j\pi\mathbb{Z} \subset 2\pi\mathbb{Z}$ for $j \geq 1$,

$$|\operatorname{sinc}(z/2^j)| \geq \frac{2 \operatorname{dist}(z, 2^j\pi\mathbb{Z})}{\pi|z|} \geq \frac{4\eta}{R_M} \quad (1 \leq j \leq J_M).$$

For $j > J_M$, $|z|2^{-j} \leq 1/4$. The power series of sinc gives $|\operatorname{sinc} u| \geq 1 - |u|^2/5 \geq e^{-|u|^2}$ for $|u| \leq 1/2$. Their product is at least

$$\exp \left[-R_M^2 \sum_{j>J_M} 4^{-j} \right] \geq e^{-1}.$$

This proves the claimed product bound. Finally, for real z , $\phi_0(z) = \mathbb{E} \cos(zX_{a^0}) \geq 1 - z^2/18$, which exceeds $1/2$ on $[0, r_0]$. $\square$

Theorem 4.2 (Finite-error recovery certificate). *Let $L \geq 1$, $M \geq 4$, and $a \in \mathcal{A}_L$. If either*

$$\operatorname{TV}(\mu_a, \mu_0) \leq \varepsilon_M(L) \quad \text{or} \quad d_K(\mu_a, \mu_0) \leq \varepsilon_M(L),$$

then

$$d_\infty(a, a^0) \leq \frac{1}{2M}. \quad (4.3)$$

Proof. By theorem 2.2, the transform difference on the circles and real segments of theorem 4.1 is at most $B_M/2$. Rouché's theorem therefore gives exactly $1 + v_2(m)$ zeros inside each circle centered at $2\pi m$. All zeros of the unknown transform are real by theorem 2.1. On the intervening real segments the unknown transform cannot vanish. On $[0, r_0]$ the same conclusion follows from the last assertion of theorem 4.1 and the measurement bound.

In the normalized coordinate $w = z/(2\pi)$, the primitive periods of the unknown factors are $\beta_j = 1/(2a_j)$ for $a_j > 0$. They satisfy all hypotheses of theorem 3.1. That lemma gives the desired bound. $\square$

Corollary 4.3 (Upper stretched-exponential rate). *For either law distance, and any fixed $L \geq 1$,*

$$\omega_{\mathcal{A}_L, D}(\varepsilon) \leq C_L \exp \left[-\sqrt{\log 2} \sqrt{\log(1/\varepsilon)} \right] \quad \text{for sufficiently small } \varepsilon. \quad (4.4)$$

Proof. The explicit threshold satisfies

$$\log \frac{1}{\varepsilon_M(L)} = \frac{(\log M)^2}{\log 2} + O_L(\log M + 1). \quad (4.5)$$

To check this, use $R_M \asymp M$ and $J_M = \log M / \log 2 + O(1)$ in (4.2). If $s = \log(1/\varepsilon)$, take $M = \lfloor \exp(\sqrt{s \log 2} - A_L) \rfloor$, with A_L sufficiently large. The upper bound implicit in (4.5) then ensures $\varepsilon \leq \varepsilon_M(L)$. Apply theorem 4.2; rounding M changes only the multiplicative constant. $\square$

4.2 Why multiple transform zeros do not destroy prefix Lipschitzness

Proof of theorem 1.4. Fix r and take $M = \max(4, 2^{r-1})$. When the law distance $\delta \leq \varepsilon_M(L)$, the proof of theorem 4.2 identifies exactly one primitive period β_j close to $d_j = 2^{j-1}$ for each $j \leq r$. Write $\Phi_a(w) = \phi_a(2\pi w)$.

Let $S_j(a)$ be the sum, with multiplicities, of the zeros in $|w - d_j| < \eta$. The progression identification in theorem 3.1 gives

$$S_j(a) = \sum_{i=1}^j 2^{j-i} \beta_i, \quad \beta_j = S_j(a) - 2S_{j-1}(a), \quad S_0(a) = 0. \quad (4.6)$$

The key is to estimate the *sum* of a multiple-root cluster before attempting to identify its individual roots.

On the contour $C_j : |w - d_j| = \eta$ we have $|\Phi_a/\Phi_0 - 1| \leq 1/2$. The principal logarithm of this ratio is well defined on a neighborhood of the contour. The argument principle and integration by parts along the closed contour yield

$$S_j(a) - S_j(a^0) = \frac{1}{2\pi i} \int_{C_j} w \left(\frac{\Phi'_a}{\Phi_a} - \frac{\Phi'_0}{\Phi_0} \right) dw \quad (4.7)$$

$$= -\frac{1}{2\pi i} \int_{C_j} \log(\Phi_a/\Phi_0) dw. \quad (4.8)$$

No logarithm inside the full disk is asserted: a branch on an annular neighborhood of the contour suffices for this identity. Using $|\log(1 + u)| \leq 2|u|$ for $|u| \leq 1/2$, we get

$$|S_j(a) - S_j(a^0)| \leq \frac{4\eta(1 + LR_M)e^{Lr_0}}{B_M} \delta. \quad (4.9)$$

The recursion (4.6) bounds $|\beta_j - d_j|$ by three times this quantity. Since $\beta_j \geq (1 - \eta)d_j$,

$$|a_j - 2^{-j}| = \frac{|\beta_j - d_j|}{2\beta_j d_j} \leq \frac{2(1 + LR_M)e^{Lr_0}}{B_M} \delta.$$

Thus one may take $\varepsilon_{L,r} = \varepsilon_M(L)$ and the displayed coefficient as $C_{L,r}$. Its logarithm is $(\log 2)r^2 + O_L(r + 1)$ by the same calculation as (4.5). $\square$

Remark 4.4 (A distinction from generic polynomial root bounds). A generic j -fold zero can split on a scale $\delta^{1/j}$. Here its constituent zeros are not arbitrary: they come from already identified harmonic progressions. Cluster sums vary linearly with the perturbation, and the recursion removes the earlier progressions. This extra structure is exactly what restores exponent one for every fixed prefix.

5 A positive-root perturbation of existing dyadic factors

5.1 A constant-coefficient perturbation with controlled real roots

Fix $0 < \tau \leq 10^{-3}$, and let $n \geq 2$. Define

$$\begin{aligned} x_j &= 16^{-j} \quad (1 \leq j \leq n), & P_n(x) &= \prod_{j=1}^n (x - x_j), \\ \Delta_n &= \tau \prod_{j=1}^n x_j = \tau 16^{-n(n+1)/2}, & Q_n &= P_n - \Delta_n. \end{aligned} \quad (5.1)$$

Only the constant coefficient changes. This elementary operation will preserve many power sums without requiring two moving reference spectra.

Lemma 5.1 (Separated positive roots). *The polynomial Q_n has exactly one simple root y_j in each interval $(3x_j/4, 5x_j/4)$. In particular $y_1 > \dots > y_n > 0$. For $\ell = n - j$, put*

$$w_\ell = (4/3)^\ell 16^{-\ell(\ell+1)/2}. \quad (5.2)$$

Then

$$\frac{|y_j - x_j|}{x_j} \leq 2\tau w_{n-j}, \quad \left| \frac{\sqrt{y_j}}{\sqrt{x_j}} - 1 \right| \leq 2\tau w_{n-j}, \quad (5.3)$$

$$|\sqrt{y_n} - \sqrt{x_n}| \geq \frac{\tau}{3} 4^{-n}. \quad (5.4)$$

The power sums satisfy

$$\sum_{j=1}^n y_j^k = \sum_{j=1}^n x_j^k \quad (1 \leq k < n), \quad \sum_{j=1}^n y_j^n - \sum_{j=1}^n x_j^n = n\Delta_n. \quad (5.5)$$

Proof. Write $q = 1/16$. On the entire interval $[3x_j/4, 5x_j/4]$, the product of distances to the preceding roots is bounded below by

$$\prod_{i < j} |x - x_i| \geq \prod_{i < j} x_i \prod_{k \geq 1} (1 - \frac{5}{4}q^k) \geq \frac{9}{10} \prod_{i < j} x_i.$$

The last inequality follows from $\prod(1 - t_k) \geq 1 - \sum t_k$ for $0 \leq t_k \leq 1$ and $(5/4) \sum_{k \geq 1} q^k = 1/12$. The distances to the following roots satisfy, with $\ell = n - j$,

$$\prod_{i > j} |x - x_i| \geq (\frac{3}{4}x_j)^\ell \prod_{k \geq 1} (1 - \frac{4}{3}q^k) \geq \frac{9}{10} (\frac{3}{4})^\ell x_j^\ell,$$

since $(4/3) \sum_{k \geq 1} q^k = 4/45$. At either endpoint of the interval, the remaining factor has magnitude $x_j/4$, so

$$|P_n(x)| \geq \frac{81}{400} (\frac{3}{4})^\ell 16^{\ell(\ell+1)/2} \prod_{i=1}^n x_i \geq \frac{81}{400} \prod_{i=1}^n x_i > \Delta_n. \quad (5.6)$$

The signs at the two endpoints are opposite. Subtracting Δ_n preserves these signs. The n intervals are disjoint, hence contain n different real roots of a polynomial of degree n . This also proves simplicity and exhausts all its roots.

At y_j , the equation $|P_n(y_j)| = \Delta_n$ and the same estimates with the j th factor omitted give

$$\frac{|y_j - x_j|}{x_j} \leq \frac{100}{81} \tau (\frac{4}{3})^\ell 16^{-\ell(\ell+1)/2} \leq 2\tau w_\ell.$$

For positive u , $|\sqrt{u} - 1| \leq |u - 1|$, proving the second relative bound. For the last root, $0 < y_n < x_i$ when $i < n$, and therefore

$$\Delta_n = |y_n - x_n| \prod_{i < n} (x_i - y_n) \leq |y_n - x_n| \prod_{i < n} x_i.$$

It follows that $|y_n - x_n| \geq \tau x_n$. Dividing by $\sqrt{y_n} + \sqrt{x_n} < 3\sqrt{x_n}$ proves (5.4). Finally, Newton's identities show that the first $n - 1$ power sums depend only on the unchanged coefficients. In degree n the constant term enters with coefficient n , giving exactly the final formula in (5.5). $\square$

5.2 The full infinite spectrum

Define $b^{(n)}$ by replacing just the first n even-indexed factors:

$$b_{2j}^{(n)} = \sqrt{y_j} \quad (1 \leq j \leq n), \quad b_i^{(n)} = 2^{-i} \quad \text{for every other } i. \quad (5.7)$$

The order of the factors does not change. Indeed, the odd factor immediately before the j th replacement is $2 \cdot 4^{-j}$, the replacement lies between $\sqrt{3/4} 4^{-j}$ and $\sqrt{5/4} 4^{-j}$, and the next odd factor is $4^{-j}/2$.

Proposition 5.2 (Size, variance, and moment matching). *The spectra (5.7) satisfy*

$$\frac{\tau}{3} 4^{-n} \leq d_\infty(b^{(n)}, a^0) \leq \|b^{(n)} - a^0\|_1 \leq 3\tau 4^{-n}. \quad (5.8)$$

Their laws have exactly the same moments as μ_0 through degree $2n - 1$, and have a different moment in degree $2n$. In particular their variance is exactly $1/9$. All their laws are supported in $[-2, 2]$, and $b^{(n)} \in \mathcal{A}_L$ eventually for every $L > 1$.

Proof. The lower bound is (5.4). For the upper bound, observe that $4^\ell w_\ell \leq (1/3)^\ell$. Thus

$$\sum_{j=1}^n |\sqrt{y_j} - 4^{-j}| \leq 2\tau 4^{-n} \sum_{\ell=0}^{n-1} 4^\ell w_\ell \leq 3\tau 4^{-n}.$$

The support assertions follow by summing the coefficients. Equation (5.5), applied to their squared lengths, preserves the full even power sums through exponent $2n - 2$. The logarithmic transform identity (2.7) then preserves cumulants and moments through $2n - 1$. Its coefficient in degree $2n$ is nonzero, and the power-sum change there is $n\Delta_n \neq 0$, so that moment is different. The variance assertion is the case $k = 1$ of (5.5). $\square$

5.3 Membership in the exact smooth class

Preserving the variance alone would not answer the question in the predecessor's class. We next check every one of its infinitely many derivative bounds, with its stated constants.

Lemma 5.3 (Derivative norms of the reference and a convolution bound). *The density f_0 is smooth and*

$$\|f_0^{(r)}\|_\infty = 2^{r(r+1)/2} \quad (r \geq 0). \quad (5.9)$$

For any summable spectrum all of whose entries are positive, its density is smooth, and

$$\|f_a^{(r)}\|_\infty \leq \frac{1}{2 \prod_{i=1}^{r+1} a_i} \quad (r \geq 0). \quad (5.10)$$

Proof. Arbitrarily many positive uniform factors imply arbitrarily fast polynomial decay of the characteristic function, and hence a C^∞ density by Fourier inversion. The density has compact support when the spectrum is summable. The first factor of a^0 has constant density one on $[-1/2, 1/2]$, and the remaining sum is supported in the same interval. Consequently $\|f_0\|_\infty = f_0(0) = 1$.

The tail after this first factor has the law of $X_{a^0}/2$. Differentiating its convolution with that first uniform gives

$$f_0'(x) = 2(f_0(2x+1) - f_0(2x-1)).$$

After r further derivatives,

$$f_0^{(r+1)}(x) = 2^{r+1}(f_0^{(r)}(2x+1) - f_0^{(r)}(2x-1)).$$

The two nonzero support interiors in this formula are disjoint; each argument ranges over the full interval $[-1, 1]$ on its respective half. Smoothness makes every boundary derivative zero. The supremum norm therefore multiplies by exactly 2^{r+1} at each step, proving (5.9).

For the general bound, differentiate each of the first r uniform densities once in the distributional sense. The resulting signed measures have total variation $1/a_i$. Keep the next uniform as a density with supremum norm $1/(2a_{r+1})$ and the rest as a probability measure. Convolution and the already established smoothness give (5.10). $\square$

Proposition 5.4 (Smoothness and geometric separation survive). *For $0 < \tau \leq 10^{-3}$, every spectrum $b^{(n)}$ belongs to $\mathcal{K}$. Given any fixed $\rho \in (1/2, 1)$, the choice*

$$0 < \tau \leq \tau_\rho := \min\left(10^{-3}, \frac{2\rho - 1}{8(2\rho + 1)}\right) \quad (5.11)$$

ensures $b^{(n)} \in \mathcal{K}_\rho$ for every $n \geq 2$.

Proof. Since $w_\ell \leq (1/12)^\ell$, the sum of all relative changes to the nontrivial coordinates is at most

$$\sum_{j=1}^n \left| \frac{b_{2j}^{(n)}}{a_{2j}^0} - 1 \right| \leq \frac{24\tau}{11} < \frac{1}{2}.$$

For every prefix, the product of its coefficient ratios is consequently at least $1 - 24\tau/11 > 1/2$. Applying (5.10),

$$\|f_{b^{(n)}}^{(r)}\|_\infty \leq \frac{1}{\prod_{i=1}^{r+1} 2^{-i}} = 2^{(r+1)(r+2)/2} = 2^{r+1} \|f_0^{(r)}\|_\infty.$$

The support and variance conditions were proved in theorem 5.2. This establishes $\mathcal{K}$ membership.

Every coordinate ratio $b_i^{(n)}/a_i^0$ lies in $[1 - 2\tau, 1 + 2\tau]$. Therefore

$$\frac{b_{i+1}^{(n)}}{b_i^{(n)}} \leq \frac{1 + 2\tau}{2(1 - 2\tau)}.$$

This is at most ρ whenever $\tau \leq (2\rho - 1)/(2(2\rho + 1))$; (5.11) is a stricter convenient choice. The index $i = 2n$, where the unchanged tail resumes, is included in the same inequality. $\square$

6 A quantitative bound against the exact reference law

Moment matching by itself is not a bound on total variation. In particular, a low-frequency estimate cannot simply be extended to all frequencies. The following argument has a common smooth background, a small head perturbation, a moment-cancelled tail, and an explicit high-frequency integral.

6.1 The unchanged odd-indexed background

The odd-indexed factors have characteristic function

$$H(t) = \prod_{\ell \geq 0} \text{sinc}\left(\frac{1}{2} 4^{-\ell} t\right). \quad (6.1)$$

For every positive integer K and $t \neq 0$,

$$|H(t)| \leq 2^{K^2} |t|^{-K}. \quad (6.2)$$

Indeed $|\text{sinc } u| \leq \min(1, 1/|u|)$ on the real axis. Apply the second bound to the first K factors and the first bound to the others. The reciprocal product of those K half-lengths is exactly 2^{K^2} .

There is also an unchanged even tail. More precisely, set

$$\Xi_n(t) = H(t) \prod_{j > n} \text{sinc}(4^{-j} t), \quad A_n(t) = \prod_{j=1}^n \text{sinc}(4^{-j} t), \quad B_n(t) = \prod_{j=1}^n \text{sinc}(\sqrt{y_j} t).$$

Then the full transform difference is

$$\phi_{b(n)}(t) - \phi_0(t) = \Xi_n(t)(B_n(t) - A_n(t)), \quad |\Xi_n(t)| \leq |H(t)|. \quad (6.3)$$

Keeping this common even tail in the identity is important, even though only its modulus bound is needed.

6.2 An explicit low-frequency estimate

For $n \geq 16$ define

$$h = \lfloor n/2 \rfloor, \quad m = \lfloor n/8 \rfloor, \quad K = \lfloor 3n/10 \rfloor, \quad T = 2^{3n/5}, \quad (6.4)$$

$$A = 5/64, \quad D_n = 2\tau h(4/3)^n 16^{-(n-h)(n-h+1)/2}, \quad (6.5)$$

$$r_n = \frac{\sqrt{5/4} 4^{-(h+1)} T}{\pi}. \quad (6.6)$$

In particular $m < n$, $K \geq 1$, and $r_n < 1$. The last fact follows from $h + 1 > n/2$, giving $r_n < (\sqrt{5/4}/\pi) 2^{-2n/5}$. Introduce

$$E_n = TD_n + \frac{2mD_n}{A} \max\left(1, \frac{AT^2}{\pi^2}\right)^m + \frac{4nr_n^{2m+2}}{(m+1)(1-r_n^2)}. \quad (6.7)$$

Lemma 6.1 (Head–tail cancellation). *For real $|t| \leq T$ one has*

$$|B_n(t) - A_n(t)| \leq E_n. \quad (6.8)$$

Proof. Split the two products after factor h . For the first h factors, theorem 5.1 implies both

$$\sum_{j \leq h} |y_j - x_j| \leq D_n, \quad \sum_{j \leq h} |\sqrt{y_j} - \sqrt{x_j}| \leq D_n. \quad (6.9)$$

To see this, use $x_j, \sqrt{x_j} \leq 1$, $n - j \geq n - h$, $(4/3)^{n-j} \leq (4/3)^n$, and the monotonicity of $\ell(\ell+1)/2$ for $\ell \geq 0$. On the real axis, the sinc function is Lipschitz with constant at most one: write it as $\mathbb{E}e^{itU}$ and use $\mathbb{E}|U| \leq 1$. Telescoping products whose factors have modulus at most one therefore bounds the head difference by TD_n .

All arguments of the tail factors have absolute value at most $\pi r_n < \pi$. Their products are positive, at most one, and admit the real logarithm (2.7). The total power sums agree through degree $n - 1$. For $1 \leq k \leq m$, the difference of the tail power sums is consequently the negative of the head difference. Since $x_j, y_j \leq A$ and $|u^k - v^k| \leq kA^{k-1}|u - v|$ on $[0, A]$, its magnitude is at most $kA^{k-1}D_n$. The contribution of these m terms to the logarithm difference is at most

$$\sum_{k=1}^m \frac{2}{k\pi^{2k}} kA^{k-1}D_n T^{2k} \leq \frac{2mD_n}{A} \max\left(1, \frac{AT^2}{\pi^2}\right)^m.$$

Here we used $\zeta(2k) < 2$.

For $k \geq m + 1$ we need no cancellation. There are at most n factors in each tail, and each squared scale is bounded by $(5/4)16^{-(h+1)}$. Thus the absolute difference of its power sums is at most twice n times that bound to the k th power. Summing the logarithmic series gives

$$4n \sum_{k \geq m+1} \frac{r_n^{2k}}{k} \leq \frac{4nr_n^{2m+2}}{(m+1)(1-r_n^2)}.$$

Since $|e^u - e^v| \leq |u - v|$ for real $u, v \leq 0$, the same sum bounds the difference of the two tail products. Combining the head and tail bounds proves (6.8). $\square$

6.3 The full-law error and its quadratic logarithm

Theorem 6.2 (Explicit anchored total-variation bound). *For $n \geq 16$ define*

$$V_n = \min \left\{ 1, \left(\frac{2TE_n^2 + 8 \cdot 2^{2K^2} T^{1-2K} / (2K-1)}{2\pi} \right)^{1/2} \right\}. \quad (6.10)$$

Then

$$\text{TV}(\mu_{b(n)}, \mu_0) \leq V_n, \quad V_n \leq \exp \left[-\frac{9 \log 2}{100} n^2 + O_\tau(n) \right]. \quad (6.11)$$

The constant implicit in $O_\tau(n)$ is independent of n .

Proof. On $[-T, T]$, theorem 6.1 and $|\Xi_n| \leq 1$ bound the squared Fourier integral by $2TE_n^2$. On its complement, $|B_n - A_n| \leq 2$, and (6.2) gives

$$\int_{|t|>T} |\phi_{b(n)}(t) - \phi_0(t)|^2 dt \leq \frac{8 \cdot 2^{2K^2} T^{1-2K}}{2K-1}.$$

Both densities are supported in $[-2, 2]$. Cauchy–Schwarz and Plancherel, with the characteristic-function convention (2.1), give

$$\text{TV}(\mu_{b(n)}, \mu_0) = \frac{1}{2} \|f_{b(n)} - f_0\|_1 \leq \|f_{b(n)} - f_0\|_2 = \left(\frac{1}{2\pi} \int_{\mathbb{R}} |\phi_{b(n)} - \phi_0|^2 dt \right)^{1/2}.$$

This proves the finite formula.

For clarity, the asymptotic exponents of every term are recorded explicitly. With $a = \log 2$, the floor functions in (6.4) give

$$\begin{aligned} \log D_n &= -\frac{1}{2}an^2 + O_\tau(n), & \log T &= \frac{3}{5}an, \\ \log r_n &= -\frac{2}{5}an + O(1), & m &= \frac{1}{8}n + O(1). \end{aligned} \quad (6.12)$$

Consequently the three positive terms of E_n are bounded respectively by

$$\exp[-\frac{1}{2}an^2 + O_\tau(n)], \quad \exp[-\frac{7}{20}an^2 + O_\tau(n)], \quad \exp[-\frac{1}{10}an^2 + O(n)]. \quad (6.13)$$

For example, the middle term adds $m \log(AT^2/\pi^2) = \frac{3}{20}an^2 + O(n)$ to $\log D_n$, while the last uses $(2m+2) \log r_n = -\frac{1}{10}an^2 + O(n)$. The low-frequency contribution after taking the square root in (6.10) is therefore at most $\exp[-\frac{1}{10}an^2 + O_\tau(n)]$. For the high-frequency contribution,

$$2K^2 \log 2 + (1-2K) \log T = -\frac{9}{50}an^2 + O(n).$$

Its square root has exponent $-\frac{9}{100}an^2 + O(n)$ and is the larger of the two bounds. Combining them proves (6.11). $\square$

Remark 6.3 (Why both parts of the Fourier argument are necessary). The first few changed factors are so small a perturbation that their contribution is quadratic-exponentially small in n . The remaining factors are small enough that a long truncated logarithmic expansion stays valid up to $T = 2^{3n/5}$. Beyond T , cancellation is no longer used: the unchanged odd-indexed factors supply the decay. This separation avoids extrapolating a moment expansion through the zeros of sinc.

7 Completion of the anchored theorem and its interpretation

Proof of theorem 1.3. The upper bound follows from theorem 4.3. For $\mathcal{K}$ and $\mathcal{K}_\rho$ use their inclusion in $\mathcal{A}_3$. In particular it yields the first inequality in (1.8).

For the lower bound fix $\tau > 0$, using (5.11) when needed. Put $c_0 = 9 \log 2 / 100$ and $s = \log(1/\varepsilon)$. By theorem 6.2, there is C_τ for which $V_n \leq \exp(-c_0 n^2 + C_\tau n)$ for all sufficiently large n . Choose

$$n = \left\lceil \frac{\sqrt{s}}{\sqrt{c_0}} + B_\tau \right\rceil, \quad (7.1)$$

where B_τ is a sufficiently large constant. Then $V_n \leq \varepsilon$. The constructed spectrum belongs to $\mathcal{K}$ and, with the smaller choice of τ , to $\mathcal{K}_\rho$. It belongs to $\mathcal{A}_L$ for sufficiently large n whenever $L > 1$. Since $d_K \leq \text{TV}$, it is admissible for both law metrics. Its parameter gap satisfies

$$\omega_{\mathcal{C},D}(\varepsilon) \geq \frac{\tau}{3} 4^{-n}.$$

Taking logarithms and using (7.1),

$$\log(1/\omega_{\mathcal{C},D}(\varepsilon)) \leq \frac{\log 4}{\sqrt{c_0}} \sqrt{s} + O_\tau(1) = \frac{20}{3} \sqrt{\log 2} \sqrt{s} + O_\tau(1).$$

This proves the final inequality of (1.8). The intermediate liminf–limsup inequality is automatic. The two-sided form (1.7) follows after slightly weakening the constants to absorb bounded terms. $\square$

Corollary 7.1 (No anchored Hölder estimate, even with lacunarity). *In each class in theorem 1.3, no constants $C < \infty$ and $\alpha > 0$ can make*

$$d_\infty(a, a^0) \leq CD(\mu_a, \mu_0)^\alpha$$

hold throughout a sufficiently small law neighborhood of μ_0 . On the other hand, for every $p > 0$,

$$\omega_{\mathcal{C},D}(\varepsilon) = o((\log(1/\varepsilon))^{-p}).$$

Proof. For the first assertion the construction gives

$$\frac{d_\infty(b^{(n)}, a^0)}{D(\mu_{b^{(n)}}, \mu_0)^\alpha} \geq \frac{\tau}{3} \exp(\alpha c_0 n^2 - O_\tau(n)) \longrightarrow \infty.$$

The second assertion follows from the upper bound because $e^{-c\sqrt{s}s^p} \rightarrow 0$ as $s \rightarrow \infty$. $\square$

7.1 Exactly which repository questions are answered

The anchored power-law question receives a negative answer in the very smooth class used by the predecessor, together with a positive quantitative replacement: the sharp exponent in the square root of the logarithm. Geometric separation with any fixed ratio strictly greater than $1/2$ does not restore a power law at the dyadic reference. The prefix question receives a positive answer under a single common support bound: every fixed prefix has exponent one, even with an unknown infinite tail.

These conclusions do not say that the full inverse has this modulus between two arbitrary nearby spectra. The special zero lattice of the exact reference is used on every contour in the upper proof. Nor do they determine the exact support endpoint of the perturbed laws: their half-support lengths tend to one, but need not equal one. Fixed variance and fixed support endpoints are different restrictions.

7.2 Why the two exponents are geometrically natural

Resolving factors down to size about $1/M$ requires viewing transform zeros at frequencies of order M . There are about $\log_2 M$ active dyadic factors there, and a conservative product lower bound loses about $\exp[-(\log M)^2/\log 2]$. This yields the square root when inverted. The construction realizes the converse scale through products of geometrically separated root distances: their logarithms are sums of arithmetic progressions, therefore quadratic in the number of factors.

This explanation is not substituted for a proof. The finite formulas (4.2) and (6.10) establish both directions without an appeal to numerical fitting. They also reveal the unresolved constant gap: the contour argument and the chosen head–tail cutoffs are not simultaneously optimized.

8 Testing complexity and honest confidence sets

8.1 A testing formulation that is genuinely informative

An estimator identically equal to a^0 has zero loss when the truth is exactly a^0 . For that reason, an assertion about “the minimax estimation rate at a single point” without a neighborhood or honesty condition is misleading. We instead consider testing a fixed null against separated alternatives.

Fix $\rho \in (1/2, 1)$. Let $N_\rho(\delta)$ be the least positive integer N for which there is a possibly randomized test $0 \leq \psi \leq 1$, based on N independent observations, with

$$\mathbb{E}_{\mu_0^{\otimes N}} \psi \leq \frac{1}{4}, \quad \sup_{\substack{a \in \mathcal{K}_\rho \\ d_\infty(a, a^0) \geq \delta}} \mathbb{E}_{\mu_a^{\otimes N}} (1 - \psi) \leq \frac{1}{4}. \quad (8.1)$$

The supremum is over the alternative, not over an arbitrarily chosen estimator at the null.

Theorem 8.1 (Quasi-polynomial sample complexity in inverse accuracy). *For fixed $\rho \in (1/2, 1)$,*

$$N_\rho(\delta) = \exp \left[\Theta_\rho(\log^2(1/\delta)) \right] \quad (\delta \downarrow 0). \quad (8.2)$$

The constants obtained here satisfy

$$\frac{9}{400 \log 2} \leq \liminf_{\delta \downarrow 0} \frac{\log N_\rho(\delta)}{\log^2(1/\delta)} \leq \limsup_{\delta \downarrow 0} \frac{\log N_\rho(\delta)}{\log^2(1/\delta)} \leq \frac{2}{\log 2}. \quad (8.3)$$

The upper test works for all alternatives in $\mathcal{A}_3$, not just $\mathcal{K}_\rho$. The same order holds with alternatives in $\mathcal{K}$ or $\mathcal{A}_L$ for fixed $L > 1$.

Proof. Let F_N be the empirical distribution function. We use the classical Dvoretzky–Kiefer–Wolfowitz–Massart bound

$$\mathbb{P}_F \left\{ \sup_x |F_N(x) - F(x)| > u \right\} \leq 2e^{-2Nu^2} \quad (u > 0). \quad (8.4)$$

Massart proved the sharp constant two; see [4] and the account in [5]. This is an external statistical input, not a new inequality claimed here.

Take $M = \max(4, \lceil 1/\delta \rceil)$ and $\varepsilon_* = \varepsilon_M(3)$. By theorem 4.2, any alternative with $d_\infty(a, a^0) \geq \delta$ has $d_K(\mu_a, \mu_0) > \varepsilon_*$, since $1/(2M) < \delta$. Reject the null when

$$\sup_x |F_N(x) - F_0(x)| > \varepsilon_*/2. \quad (8.5)$$

Under the null, (8.4) bounds the rejection probability by $2 \exp(-N\varepsilon_*^2/2)$. Under any alternative, acceptance implies an empirical CDF error larger than $\varepsilon_*/2$ relative to its own law, and the same bound applies. Thus a finite upper bound is

$$N_\rho(\delta) \leq \left\lceil \frac{2 \log 8}{\varepsilon_M(3)^2} \right\rceil. \quad (8.6)$$

Equation (4.5) proves the upper logarithmic constant in (8.3).

For the lower bound choose τ as in (5.11) and, for small δ , put

$$n = \left\lfloor \frac{\log(\tau/(3\delta))}{\log 4} \right\rfloor.$$

Then $n \geq 16$ eventually and $d_\infty(b^{(n)}, a^0) \geq \delta$. The elementary product-measure bound gives

$$\text{TV}(\mu_{b^{(n)}}^{\otimes N}, \mu_0^{\otimes N}) \leq N \text{TV}(\mu_{b^{(n)}}, \mu_0) \leq NV_n.$$

For any test, its two error probabilities at these two laws sum to at least one minus this total variation. If $N \leq 1/(4V_n)$, their sum is at least $3/4$, which is incompatible with both being at most $1/4$. Therefore $N_\rho(\delta) > \lfloor 1/(4V_n) \rfloor$. Using (6.11) and $n = \log(1/\delta)/\log 4 + O_\rho(1)$ yields

$$\log N_\rho(\delta) \geq \frac{9}{400 \log 2} \log^2(1/\delta) - O_\rho(\log(1/\delta)).$$

The same construction is available in $\mathcal{K}$ and eventually in $\mathcal{A}_L$ for any $L > 1$; in the latter case the upper test uses $\varepsilon_M(L)$. $\square$

8.2 Confidence sets that shrink at the reference

Fix $L \geq 1$ and a coverage error $0 < \alpha < 1$. Define

$$t_N = \sqrt{\frac{\log(2/\alpha)}{2N}}, \quad \mathcal{C}_N = \{a \in \mathcal{A}_L : \sup_x |F_a(x) - F_N(x)| \leq t_N\}. \quad (8.7)$$

For every $a \in \mathcal{A}_L$, (8.4) gives $\mathbb{P}_a\{a \in \mathcal{C}_N\} \geq 1 - \alpha$. At the reference law, on an event of that probability, every candidate in $\mathcal{C}_N$ satisfies $d_K(\mu_a, \mu_0) \leq 2t_N$. Hence

$$\sup_{a \in \mathcal{C}_N} d_\infty(a, a^0) \leq \omega_{\mathcal{A}_L, d_K}(2t_N) \leq C_{L, \alpha} \exp \left[-\sqrt{\frac{\log 2}{2}} \sqrt{\log N} \right] \quad (8.8)$$

for sufficiently large N . For each fixed prefix, theorem 1.4 instead gives a radius at most $2C_{L, r} t_N = O_{L, r, \alpha}(N^{-1/2})$.

These are honest coverage statements over $\mathcal{A}_L$ with contraction at the reference. They are not claimed as a globally optimal estimator, nor as a finite algorithm for representing the infinite-dimensional confidence set. Numerically implementing (8.5) also requires validated values of F_0 ; its statistical proof assumes that reference CDF is known exactly. A uniform CDF approximation error can be included by enlarging the threshold, but an implementation and its numerical-error budget are separate tasks.

9 Reproducible checks and finite-error scales

9.1 Exact arithmetic, diagnostics, and what each establishes

The accompanying program `code/verify.py` uses Python's rational `Fraction` arithmetic for the finite polynomial checks. Its default choice is $\tau = 1/1000$. On the run included with this article, it completed the following checks without an assertion failure:

Exact check	Number
Opposite endpoint signs for the perturbed root intervals, $2 \leq n \leq 20$	209
Unchanged power sums below degree n , $2 \leq n \leq 20$	190
First changed power sum equals $n\Delta_n$	19
Only the constant polynomial coefficient changes	19
Divisor inversion returns the power-of-two indicator, $1 \leq m \leq 512$	512
Total finite exact checks	949

These are regression checks, not a proof for all n or a formal proof of the analytic theorems. The universal proofs are theorems 3.1, 5.1 and 6.2.

A separate part of the program uses `mpmath` for high-precision root bisection at $n = 2, 4, 6, 8$ and for evaluations of the explicit analytic bounds. The root calculations used 100, 100, 158, and 242 decimal digits, respectively. These values are floating-point diagnostics, not interval certificates. The exact rational endpoint signs and the proved isolation lemma are kept logically separate from the floating-point residuals. The recorded environment was Python 3.13.5 and `mpmath` 1.3.0.

9.2 Scale of the anchored witnesses

n	Gap lower bound $(\tau/3)4^{-n}$	TV upper bound V_n
16	7.7610×10^{-14}	2.1467×10^{-6}
32	1.8070×10^{-23}	4.9268×10^{-26}
64	9.7958×10^{-43}	1.2260×10^{-106}
128	2.8787×10^{-81}	6.8525×10^{-434}
256	2.4861×10^{-158}	5.3403×10^{-1754}
512	1.8542×10^{-312}	9.2947×10^{-7058}

Table 1: Rounded evaluations of the proved formulas, with $\tau = 1/1000$. The second column is a lower bound, not a measured spectral distance. The third is an analytic upper bound, not numerical integration of an actual TV distance. The rounding displayed is not directed interval rounding.

The construction is intentionally conservative. At small n , the Fourier upper bound can be much larger than the spectral gap. The asymptotic obstruction emerges from the quadratic logarithm of the law bound versus the linear logarithm of the spectral gap. For example, the value of $-\log V_n/n^2$ at $n = 512$ is approximately 0.06199, approaching the proved upper-bound exponent $9 \log 2/100 \approx 0.06238$. This concerns V_n alone; it does not identify the actual TV asymptotics.

9.3 Scale of a finite recovery certificate

The upper certificate does not recover roots by numerically searching a noisy transform. It states an implication: a proved measurement bound below the threshold forces the correct finite divisor pattern and hence the displayed parameter accuracy. A practical solver would need a representation of the observed transform, validated contour bounds, and an error budget for each numerical operation.

M	J_M	$\log_{10} \varepsilon_M(3)$	Guaranteed spectral error $1/(2M)$
4	7	-18.7738	1.2500×10^{-1}
16	9	-28.9611	3.1250×10^{-2}
64	11	-41.5784	7.8125×10^{-3}
256	13	-56.6109	1.9531×10^{-3}
1024	15	-74.0537	4.8828×10^{-4}
4096	17	-93.9054	1.2207×10^{-4}

Table 2: Rounded evaluations of (4.2) for the support bound $L = 3$. These sufficient thresholds deliberately favor a transparent uniform proof over good practical constants.

9.4 How to reproduce the package

Run `python code/verify.py` from the package root, or invoke the script by its absolute path from another directory. It writes `verification.json`, `verification.txt`, `analytic_bounds.csv`, and `rouche_thresholds.csv` to `data-rerun/`, or to the directory given by `--output-dir`; the recorded copies are in `data/`. The exact checks use only the Python standard library; the diagnostic portion needs `mpmath`. The TeX source contains the displayed tables directly, so rebuilding the article does not require Python. Two passes of `pdflatex article.tex` resolve its references; `latexmk -pdf article.tex` is also suitable. No shell escape or network access is needed to compile it.

10 Proof dependencies and repository provenance

10.1 Claim-level dependency audit

Claim	Inputs and scope
Finite upper certificate	Entire sinc products, the explicit contour bound, Rouché’s theorem, and the stable divisor lemma. No separation assumption on the unknown spectrum.
Prefix Lipschitz theorem	The same finite zero counts plus the contour sum-of-zeros identity. A logarithm is used only near the contour, not across zeros in its interior.
Anchored counterexamples	Exact constant-coefficient perturbation, isolated positive roots, Newton identities, and the unchanged odd-indexed factors.
Smooth-class membership	Exact reference derivative norms and the product bound for every prefix, as well as exact variance matching.
TV bound	A real logarithm only below the first tail zero, and an independent high-frequency product bound followed by Plancherel.
Sharp exponent	The upper certificate and the explicit lower witnesses, with integer rounding handled in both conversions.
Testing result	The preceding deterministic results, product-TV contraction, and the external DKW–Massart inequality.

No mathematical conclusion is inferred solely from finite numerical tests. No theorem from an unreviewed repository research article is imported as an assumption. Conversely, use of standard analysis is explicit: this manuscript does not claim a new proof of Rouché, Plancherel, or the DKW inequality.

10.2 Inspected source and exact question mapping

The principal predecessor at the pinned commit is

```
Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/
drafts/inverse-and-sampling/Recovering_Uniform_Factors_Fabius_Rvachev/
article.tex.
```

Its content blob identifier is

```
90395cdfd50b8d8e191d2753616fc47c890a3311.
```

The repository snapshot is `39472967ce67566d7c530fd5b8d6a3a95e68fe22`. It was read through the GitHub connector, including the opening definition of $\mathcal{K}$ and the final research-question section. The following titles specify the comparison more robustly than a line number alone:

Predecessor question	Result here
Pointwise recovery of the exact dyadic spectrum	Theorems 1.3 and 7.1: sharp stretched-exponential exponent and no Hölder estimate in the stated smooth class.
Geometrically separated classes	Theorems 1.3 and 5.4: negative power-law answer at the dyadic anchor for every fixed ratio $\rho > 1/2$; not a full pairwise classification.
Fixed leading factors versus the complete sequence	Theorem 1.4: local Lipschitz recovery of every fixed prefix with only an ℓ^1 support bound on the unknown full spectrum.
Matching statistical upper bounds	Theorem 8.1 and (8.8): matching-order anchored testing and honest confidence contraction, not the predecessor's global minimax problem.

Editorial note (ProveIt, 2026-09-29). Lean crosswalk, not cited by the article. The zero multiplicity $c_m = 1 + v_2(m)$ of (2.4) is the machine-checked theorem `Fabius.analyticOrderAt_rvachevFourierProduct_int` in `Analysis/FabiusFunction/Lean/FabiusFunction/IntegerZeroAnalyticOrder.lean`. That $1 + v_2(m) \geq r + 1$ exactly when $2^r \mid m$, the divisor count behind theorem 3.1, is `Fabius.dyadicZeroMultiplicity_ge_succ_iff_pow_two_dvd` in `DyadicZeroMultiplicity.lean` of the same directory. The derivative norms (5.9) are `Fabius.isGreatest_abs_iteratedDeriv_rvachevUp` (`GlobalBounds.lean`), and the variance $1/9$ is `Fabius.integral_sq_mul_rvachev_eq_one_ninth` (`DyadicSpecializations.lean`). Exact identifiability of dyadic exponent sequences from zero orders, a noise-free counterpart of the divisor inversion, is proved in `GeneralizedRvachevIdentifiability.lean`. No quantitative statement of this article is formalized.

The adjacent inverse-and-sampling README was also inspected. It lists the finite-factor, Gaussian-confounding, and flat-boundary recovery articles as separate recent arrivals. Their fixed-capacity questions are different from the present unknown-infinite-tail prefix result. This was a targeted source review, not an exhaustive audit of every document in ProveIt or every paper in the literature. No repository file was modified.

10.3 A concrete formalization route

A natural first module is the finite divisor-count lemma: encode the observed zero incidences as a finite multiset, prove the no-phase-slip implication, and solve the triangular divisor system. This module does not need probability or Fabius analysis.

A second algebraic module can formalize the geometric root intervals, constant-coefficient perturbation, Newton identities, and the prefix product bound. Rational endpoint inequalities

offer small finite certificate instances, while the general estimates need elementary ordered-field and geometric-series lemmas. Positivity and preservation of sorted rank must remain explicit.

The analytic modules should then separate three interfaces: compactly supported law distance to contour error; entire-product zeros and argument-principle counts; and the Fourier head–tail estimate with its L^2 -to-TV conversion. Finally, metric corollaries and testing statements can be assembled from these interfaces. The sum-of-zeros proof should not be replaced by a generic root continuity theorem, since that would lose the prefix exponent one.

This is a dependency plan, not a claim that a particular current mathlib API already exposes every required theorem. No Lean build was run, and no uncompiled Lean skeleton is included as apparent evidence of formal verification.

11 Further research questions

Research question 1 (The exact stretched-exponential constant). Does

$$\lim_{\varepsilon \downarrow 0} \frac{\log(1/\omega_{\mathcal{C},D}(\varepsilon))}{\sqrt{\log(1/\varepsilon)}}$$

exist, and how does it depend on the class and law metric? The present interval $[\sqrt{\log 2}, (20/3)\sqrt{\log 2}]$ is wide. Optimizing the retained background, the root spacing of the replaced subsequence, and the Fourier cutoffs is one route to a better lower construction. A sharp contour product bound, rather than a common bound for every early factor, may improve the upper constant independently.

Research question 2 (Pairwise recovery on geometrically separated classes). Determine the optimal pairwise modulus when both unknown spectra satisfy $a_{j+1} \leq \rho a_j$. The present upper argument fixes one transform to the exact dyadic zero lattice; it does not give a pairwise theorem uniformly on a neighborhood. Geometric separation removes the predecessor’s dense blocks but does not remove the anchored counterexamples constructed here. A positive theorem must therefore have a modulus no better than the anchored stretched-exponential obstruction.

Research question 3 (The sharp separation boundary $\rho = 1/2$). What happens if $a_{j+1} \leq a_j/2$ exactly, with variance fixed at $1/9$? Our perturbations require slack above $1/2$. At equality, adjacent one-sided constraints may rule out the compensating scale changes that preserve many power sums. The case $\rho < 1/2$ does not contain the dyadic reference at all, so it is a different anchored problem rather than a strengthening of the same one.

Editorial note (ProveIt, 2026-09-30). Answered by `../Lacunarity_Boundary_Geometric_Uniform_Spectra/` (batch 68 of the repository’s `docs/incoming/`), written against this article and in its normalization (U_j uniform on $[-1, 1]$, variance $1/9$, so $\sum_j a_j^2 = 1/3$). If $a_{j+1} \leq a_j/2$ for every j , then

$$\sup_j |a_j - 2^{-j}|^2 \leq \frac{75}{4} \left(\frac{19}{675} - \mathbb{E}X_a^4 \right),$$

and the fourth-moment deficit vanishes only at a^0 . Hence the anchored modulus at $\rho = 1/2$ is $\Theta(\sqrt{\varepsilon})$, a power law, in total variation and in Kolmogorov distance, with matching deleted-factor witnesses in $\mathcal{K}$ whose ratios are at most $1/2$; the stretched-exponential scale of theorem 1.3 does not persist at the boundary. At $\rho = 1/2$ it also answers the next question negatively (there $\sum_j a_j \leq 1$, with equality only at a^0 , so no witness with $\sum_j b_j = 1$ exists), and the question on prefix constants inside its critical cone only (local Lipschitz constants $\Theta(2^r)$). The pairwise modulus, the exact-support question with slack, and the prefix constants on $\mathcal{A}_L$ remain open.

Research question 4 (Exact support together with exact variance). Can the lower witnesses be chosen with $\sum_j b_j = 1$ exactly, while retaining variance $1/9$, geometric separation, and the sharp logarithmic exponent? The construction here controls the sum to within $3\tau 4^{-n}$ but does not fix it. A two-parameter polynomial perturbation may enforce the extra square-root sum, yet its effect on low-order cumulants and on the quadratic Fourier exponent needs proof.

Research question 5 (Sharp growth of prefix conditioning constants). Find the optimal dependence on r of the local Lipschitz constant and the radius on which it applies. We prove an upper bound $\log C_{L,r} \leq (\log 2)r^2 + O_L(r)$, but not a matching lower coefficient. This is the missing quantitative transition between root- N confidence for a fixed prefix and stretched-logarithmic contraction for the complete spectrum. Allowing $r = r(N)$ to grow is an especially concrete formulation.

Research question 6 (Other arithmetic geometric references). For a reference $a_j = cq^j$, when does the full inverse have the same square-root logarithmic exponent, and when are fixed prefixes Lipschitz? If q is the reciprocal of an integer, the zero multiplicities retain a divisibility structure, suggesting an analogue of theorem 3.1. For a general q , the harmonics can be incommensurate and nearly collide. A proof must quantify those near coincidences rather than assume the dyadic counting argument survives unchanged.

Research question 7 (Square-summable spectra and Gaussian confounding). Extend quantitative anchored recovery to the parameter space allowing square-summable but nonsummable lengths and an unknown Gaussian component, as in the qualitative theory of Billey–Swanson [2]. Compact support is used decisively to turn Kolmogorov error into entire-transform error on contours. Which tail or exponential-moment hypothesis can replace it, and which parameter metric sees escaped Gaussian variance correctly?

Editorial note (ProveIt, 2026-09-29). The predecessor’s corresponding question, “Gaussian components and square-summable spectra”, is answered for the Gaussian-variance functional by the repository article `inverse-and-sampling/Gaussian_Dust_Christoffel_Recovery_Uniform_Factors/` (filed 2026-09-29, after the snapshot this article read; unreviewed), in a model with a known smooth compactly supported background: without a tail restriction the exact minimax risk for the variance is $V/2$ at every sample size. Quantitative recovery of the scales themselves in that space, anchored or pairwise, is not treated there.

Research question 8 (Validated inversion and proof-carrying certificates). Build a finite-prefix solver that outputs both candidate lengths and certificates for the contour inequalities and divisor counts. A useful first milestone is a checker for a fixed M with rational enclosure data and a proved tail estimate for the reference sinc product. The statistical CDF error and the numerical transform error must be tracked separately. The explicit thresholds in (4.2) are a correctness baseline, not a recommendation to use their conservative constants unchanged.

Research question 9 (Adaptive confidence away from the exact anchor). Characterize reference spectra for which honest confidence sets have root- N contraction for every fixed prefix, and quantify the whole-spectrum contraction across different references. The confidence construction (8.7) is honest globally but its proved contraction rate here is anchored. Uniform adaptation would require a comparison theorem between local zero geometry and the inverse modulus, not merely the pointwise analysis of μ_0 .

12 Conclusion

Exact identifiability of a uniform-convolution spectrum conceals several distinct quantitative regimes. At the dyadic Rvachev law, the entire infinite spectrum has an anchored modulus

with the sharp scale $\exp[-\Theta(\sqrt{\log(1/\varepsilon)})]$. It admits no positive Hölder exponent even under fixed variance, the predecessor’s uniform bounds on all derivatives, and geometric separation arbitrarily close to the dyadic ratio. Yet every fixed leading prefix is locally Lipschitz when the unknown tail is constrained only by a common support bound.

Both statements arise from the special structure of uniform factors. Harmonic zero progressions allow stable divisor reconstruction and stable cluster centroids; a constant-coefficient perturbation of separated squared scales hides a moving tail behind quadratically small logarithmic law error. The explicit certificates, statistical consequences, and finite checks make these proposed results suitable for independent review and a staged formal verification effort. The exact leading constants, the ratio-1/2 boundary, and pairwise stability on separated classes remain distinct next targets.

References

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